

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2152

Chattanooga Psc 100 kV
April 27, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2152. The Interconnection Customer proposes a 62.0 MW solar photovoltaic facility interconnecting at the Chattanooga Psc 100 kV substation in the Michigan area. The request was received on August 05, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 543 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,145,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Chattanooga Psc / South Cleveland 100kV	Almena / Portola 100kV	116.7	149	100.0
Almena / Portola 100kV L	Oceanside / Palisades 500kV	108.8	185	49.0
S Cleveland / Imperial Valley 100kV	Portola Substation / Almena 100kV Line	105.0	157	100.0
Bushkill / Portola 100kV line	Ganesha / Portola Substation 100kV	104.1	107	37.8
Ganesha / Weeds Lake 100kV	Reading / Hogan 100kV line	103.5	117	30.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
South Cleveland Substation / Chattanooga Psc 100kV	Rebuild line with higher-capacity conductor	\$5,462,000	34% (\$1,828,000)
Almena / Portola 100kV line	Reconductor line	\$480,000	17% (\$83,000)
S Cleveland / Imperial Valley 100kV	Reconductor line	\$2,909,000	34% (\$973,000)
Bushkill / Portola 100kV line	Reconductor line	\$3,621,000	22% (\$816,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ganesha / Weeds Lake 100kV	Reconductor line	\$3,405,000	13% (\$445,000)
	Total allocated to project		\$4,145,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.